

DO NOT REMOVE FROM EXAM VENUE

University of Plymouth

MODULE CODE: ELEC345

TITLE OF PAPER: High Speed Communications

TIME ALLOWED Three hours

DATE Wednesday 15th January 2020

TIME 9:00 AM - 12:00 PM

FACULTY Science and Engineering

SCHOOL Engineering, Computing and Mathematics

ACADEMIC YEAR 2019/20

STAGE Three

INSTRUCTIONS TO CANDIDATES:

Candidates should attempt any **FOUR** questions from the **FIVE** available.
All questions carry equal marks. Marks for parts of questions are shown where appropriate.

Data provided:

Tables

Equation list

Four full size Smith Charts – please attach to answer book.

Candidates are not permitted to look at the examination paper until instructed to do so.

1. This question is about propagation and link budgets.
- (a) Using the free space path loss equation calculate the losses that will be seen at the edge of a 1km cell at the approximate frequencies used in 3G (900 MHz and 1800 MHz) and 4 G (2100 MHz). Show all your workings and state any assumptions.
- [5 marks]
- (b) GPS satellites orbit at an altitude of 21000 km, have a transmit power of 25.6 W, antenna gain of 13 dBi and transmit on 1575 MHz. The end of life for the system comes when the received power drops below 160 dBW:
- (i) Calculate the received signal power (on campus) showing all workings and stating any assumptions.
- [5 marks]
- (ii) What margin is there until end of life for the system?
- [1 mark]
- (iii) Why might you receive different signal strengths from each satellite?
- [1 mark]

Question 1 continued ...

Question 1 continued

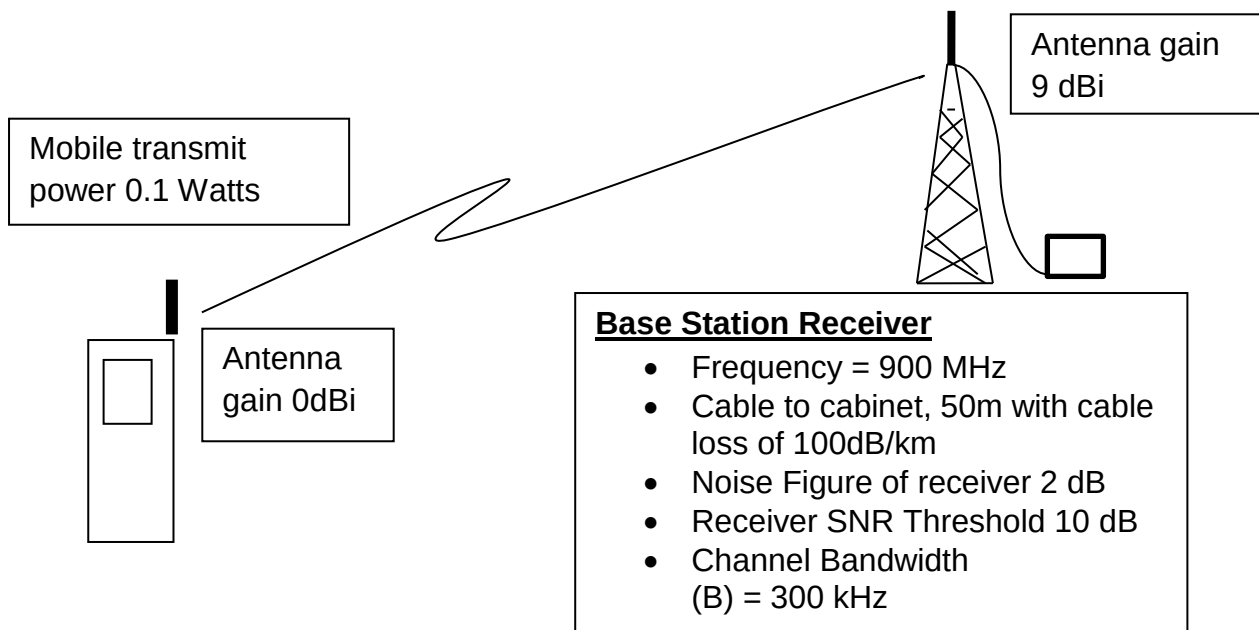


Figure Q1 Cellular example

(c) Given the scenario in Figure Q1 with 500 m between the mobile and base station:

(i) What is the signal to noise ratio at the receiver and is it above the threshold for operation? Show all workings and state assumptions.

[8 marks]

(ii) What is the maximum distance before the SNR drops below the working threshold?

[5 marks]

2. This question is about transmission lines and L matching.

(a) A 15-volt source with impedance $R_s = 25\Omega$ is connected to a transmission line with $Z_0 = 50\Omega$, length = 0.1 m and $V_p = 1 \times 10^8$ m/s followed by a load $Z_L = 100\Omega$.

(i) Sketch the circuit layout showing the source, transmission line and load and calculate the voltage going onto the line, and the voltage the line will settle to/charge to.

[5 marks]

(ii) Calculate the reflection coefficients and voltage reflections to find the time it takes for the line to get within 1% of the final charge value. Sketch a bounce diagram of these.

[8 marks]

(b) Using a Smith chart design a L-match to connect the 25Ω input to the 50Ω transmission line and calculate the component values required if the 15-volt source was swapped for a 200 MHz signal.

[12 marks]

3. This question is about the radio environment and diversity techniques.
- (a) List all the four cellular medium access techniques covered in the module; briefly explain how they operate.
[5 marks]
- (b) Along with Angle diversity what are the other diversity techniques and why are they necessary?
[10 marks]
- (c) Given received signal strengths of 4 dB, 6 dB, 7 dB and 10 dB and assuming a threshold of 2 dB is required for error free reception.
- (i) List the diversity combination techniques and their outputs in dB.
- (ii) Repeat but with the 10 dB signal having dropped to 1 dB.
[10 marks]

4. This question is on stub matching.

(a) A load impedance, $Z_L = 70 - j80\Omega$ is connected to a 50Ω transmission line. Either by calculation or using the smith chart provided find:

- (i) The voltage reflection coefficient.
- (ii) The voltage standing wave ratio in both linear form and dBs.
- (ii) The return loss and the power reflected if power in was 1 mW.

[7 marks]

(b) For the load, $Z_L = 70 - j80\Omega$ use a series stub to match the load to a 50Ω transmission line. Choose your terminations to keep all lengths as short as possible and state distances in wavelengths.

[8 marks]

(c) Redo the match in (b) using a parallel, double stub for which the spacing between stubs is $3/8\lambda$ and the spacing between the first stub and the load is 0.053λ .

[10 marks]

5. This question is on filter design.

- (a) Show the general form of a first, second and third order Butterworth filter transfer functions on the same sketch labelling or explaining the differences. (Sketch the Bode plot/frequency response.)

[5 marks]

- (b) Using the tables provided design a passive ladder network to realise a Butterworth filter with $F_c = 20$ MHz and $F_s = 60$ MHz to achieve $A_{\min} = 40$ dB between a 50Ω source and load.

[10 marks]

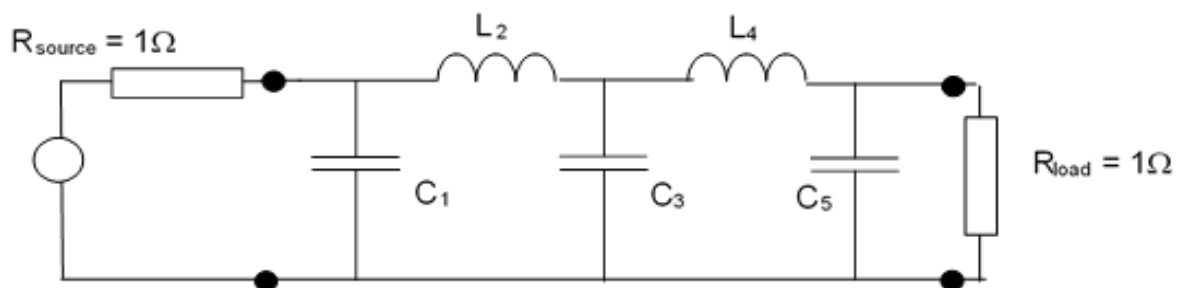
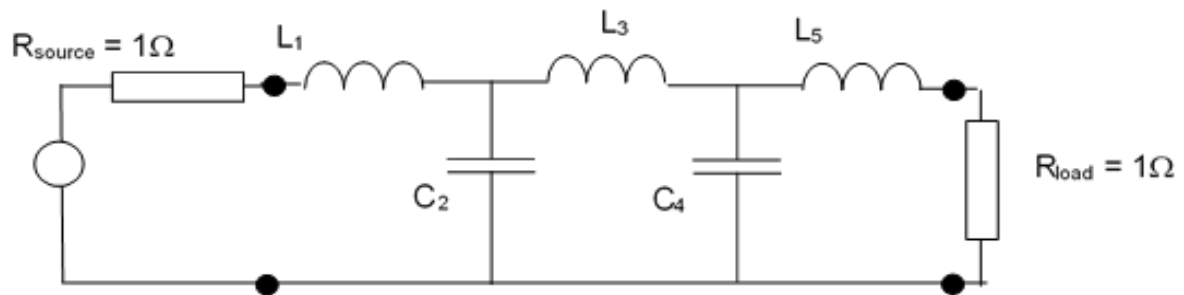
- (c) Use the attached filter tables to design an active high pass filter with $F_s = 20$ MHz and $F_c = 60$ MHz to achieve $A_{\min} = 40$ dB scaling with $10\text{ k}\Omega$ resistors.

[10 marks]

FILTER DESIGN TABLES

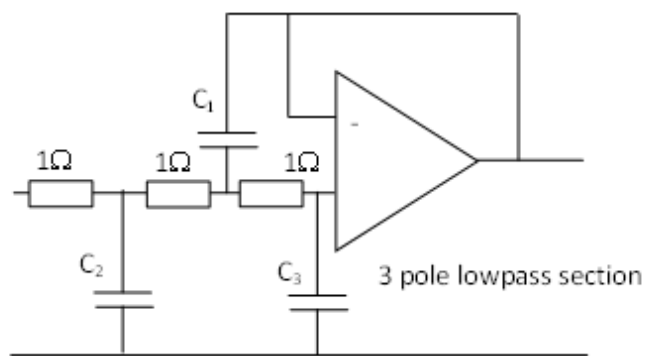
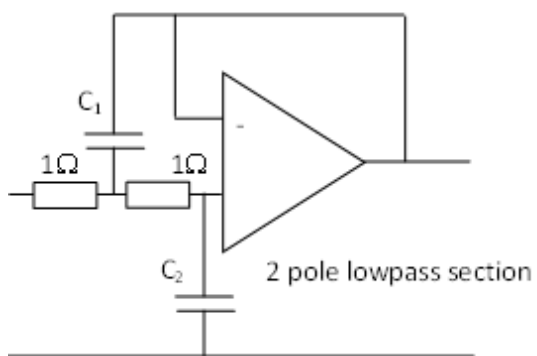
Normalised Butterworth ladder network element values

n	C ₁	L ₂	C ₃	L ₄	C ₅	L ₆	C ₇
2	1.4142	1.4142					
3	1	2	1				
4	0.7654	1.8478	1.8478	0.7654			
5	0.618	1.618	2	1.618	0.618		
6	0.5176	1.4142	1.9319	1.9319	1.4142	0.5176	
7	0.445	1.247	1.8019	2	1.8019	1.247	0.445
	L ₁	C ₂	L ₃	C ₄	L ₅	C ₆	L ₇

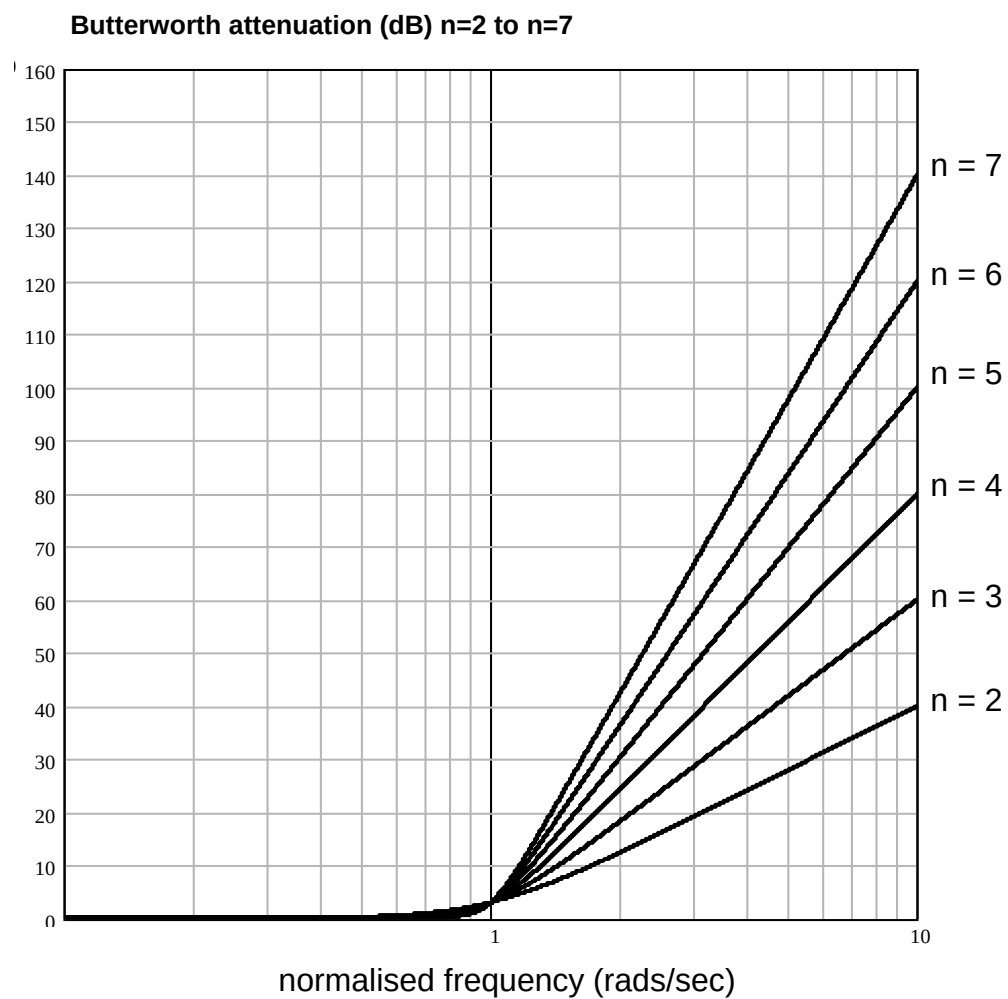
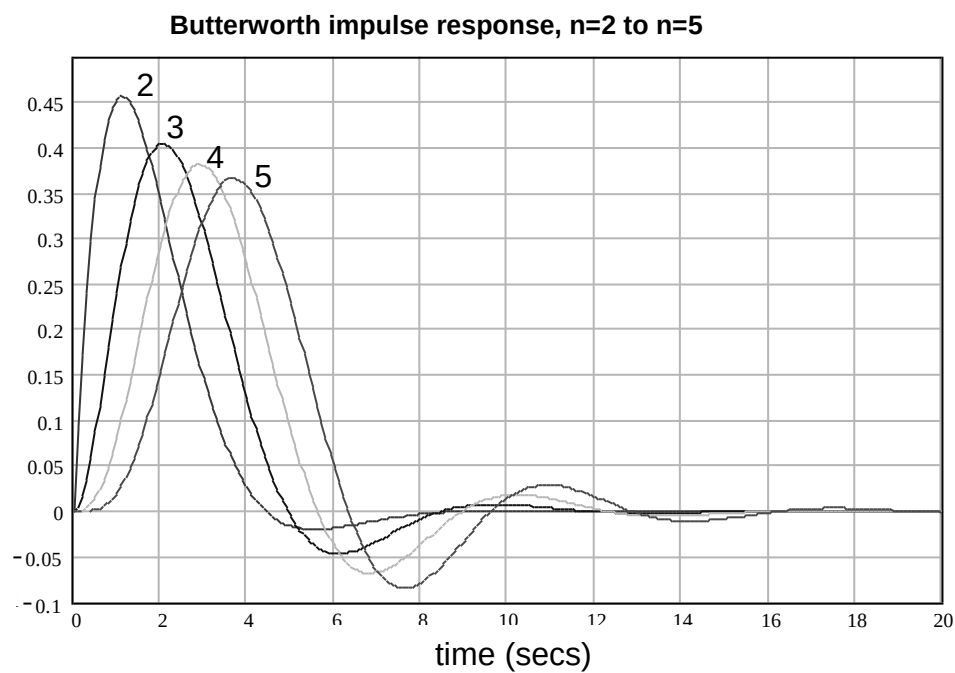


Normalised Butterworth active element values

n	C₁	C₂	C₃
2	1.414	0.7071	
3	3.546	1.392	0.2024
4	1.082 2.613	0.9241 0.309	
5	1.753 3.253	1.354 0.309	0.4214
6	1.035 1.414 3.863	0.966 0.7071 0.2588	
7	1.531 1.604 4.493	1.336 0.6235 0.2225	0.4885



Normalised Butterworth impulse response and attenuation characteristics



Useful Equations

$$L \Rightarrow \frac{LR_o}{2\pi F_c} \quad C \Rightarrow \frac{C}{2\pi F_c R_o} \quad R \Rightarrow R.R_o \quad A_s = \frac{F_s}{F_c}$$

$$R_{HP} = \frac{1}{C_{LP}} \quad \text{and} \quad C_{HP} = \frac{1}{R_{LP}}$$

$$L = 20 \log_{10} \left(\frac{4\pi R}{\lambda} \right) \text{ dB}$$

$$C = P_T + G_T - PEPL + G_R - \beta \quad [\text{dBm}]$$

$$PEPL = 20 \log_{10} \left(\frac{R^2}{h_T h_R} \right) \quad [\text{dB}] \quad (\text{plane earth path loss})$$

$$\beta = 20 + \frac{f_{\text{MHz}}}{40} + 0.18L - 0.34H + K \quad [\text{dB}]$$

$$K = \begin{cases} 0.094U - 5.9, & \text{for inner city areas} \\ 0 & , \text{elsewhere} \end{cases}$$

$$\left(f_d = \frac{v}{\lambda} = \frac{v}{c} f_{\text{Hz}} \right)$$

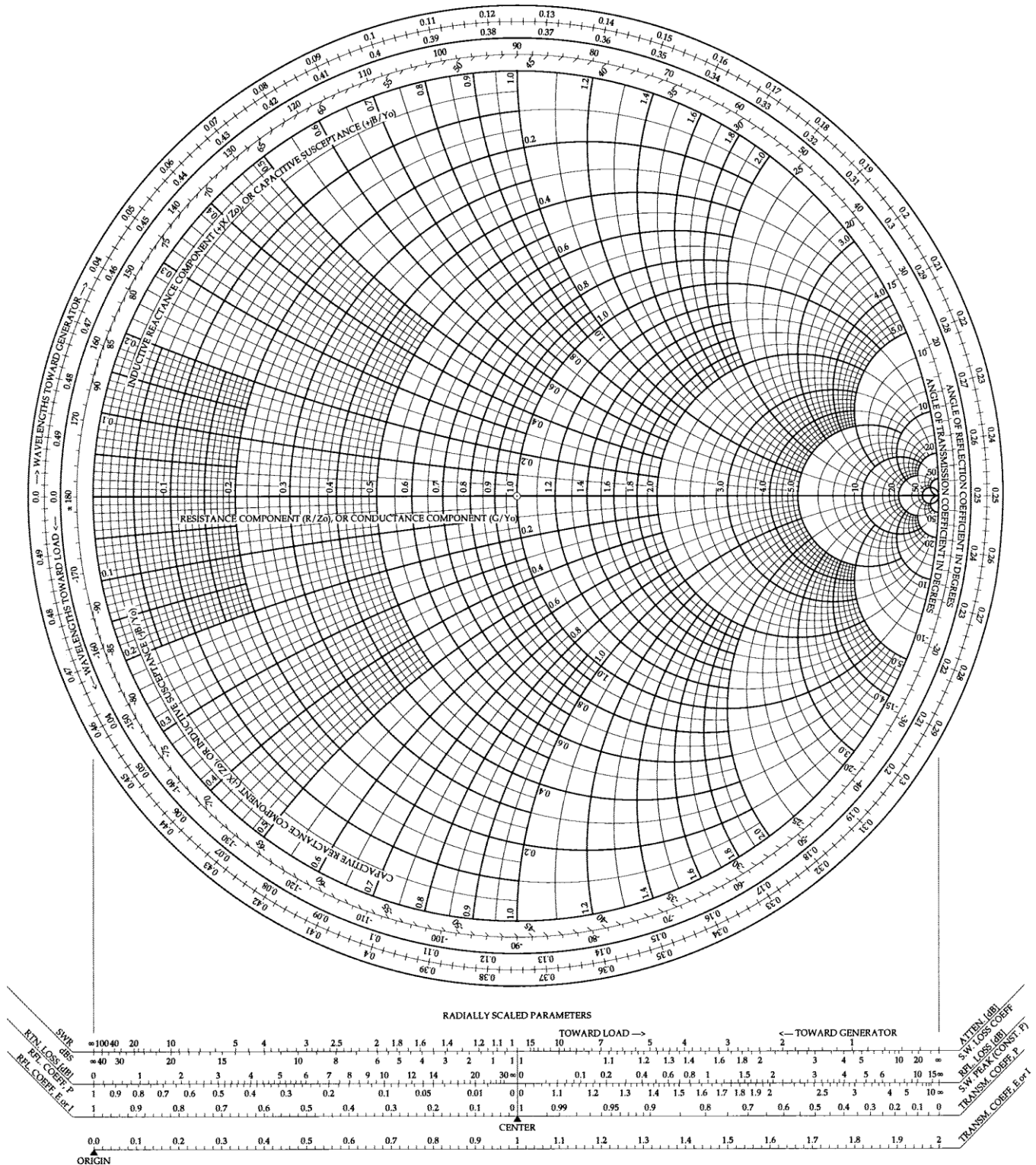
$$\gamma_{\text{div},2} = \frac{\gamma_1 + \gamma_2 + 2\sqrt{\gamma_1 \gamma_2}}{2}$$

$$\gamma_{\text{div},N} = \sum_{i=1}^N \gamma_i$$

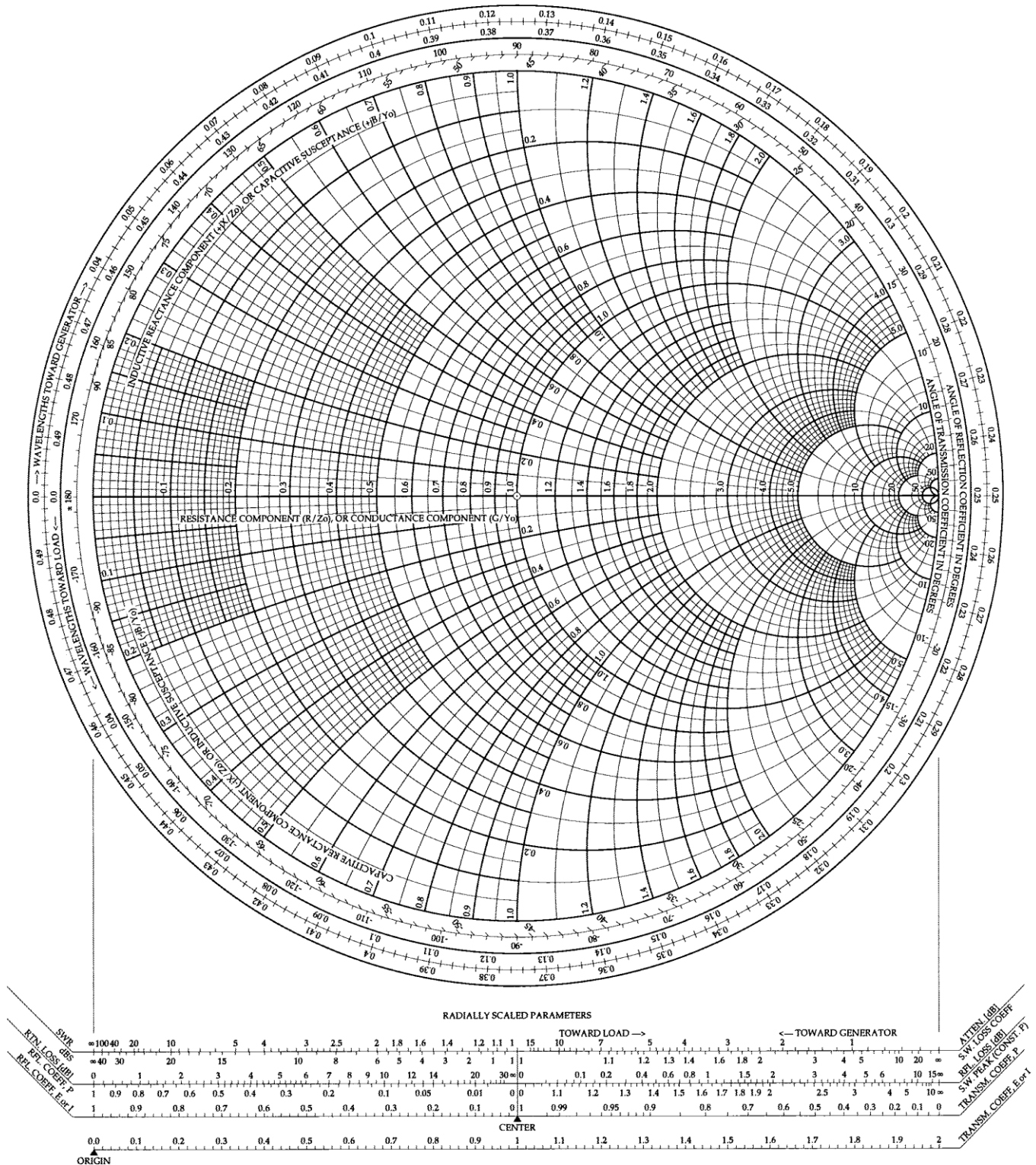
$$N = -204 + F + 10 \log_{10}(B) \quad (\text{dBW})$$

$$\Gamma = \frac{Z - Z_0}{Z + Z_0} \quad \text{return loss} = 10 \log P_r / P_i$$

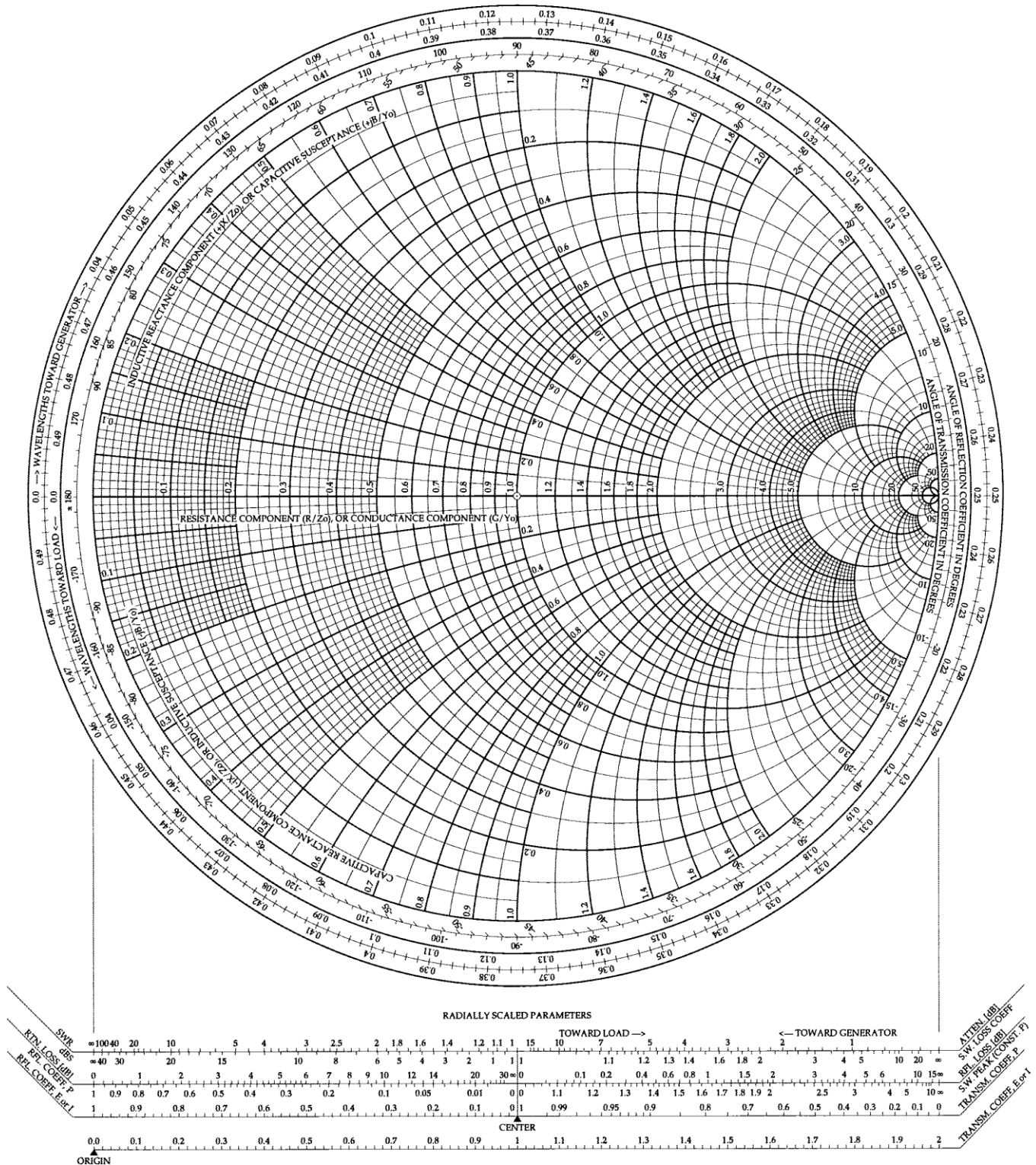
The Complete Smith Chart



The Complete Smith Chart



The Complete Smith Chart



The Complete Smith Chart

